

RegimeShift

Causal market-regime allocation research

Quantitative research case study | 2026

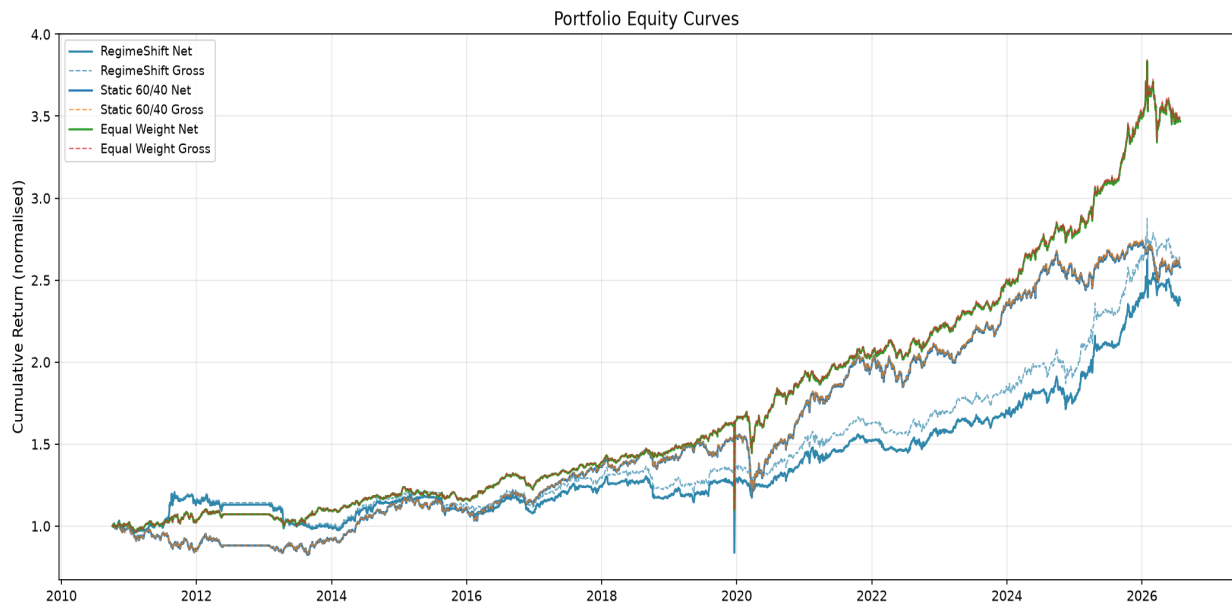
This report is an honest retrospective study, not an investment recommendation or a claim of proven alpha.

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Executive summary

Under causal NEXT_CLOSE execution and a base 10 bps one-way asset-level cost assumption, RegimeShift underperformed both static benchmarks on headline risk-adjusted performance. The project makes that negative result visible and tests whether the model complexity is justified.



Research question

Can a three-state market-regime allocation improve fixed Indian multi-asset allocations after causal timing, common-horizon benchmark alignment, and stated costs? Parameters were frozen; no performance-driven tuning was performed.

Data and index-level tradability

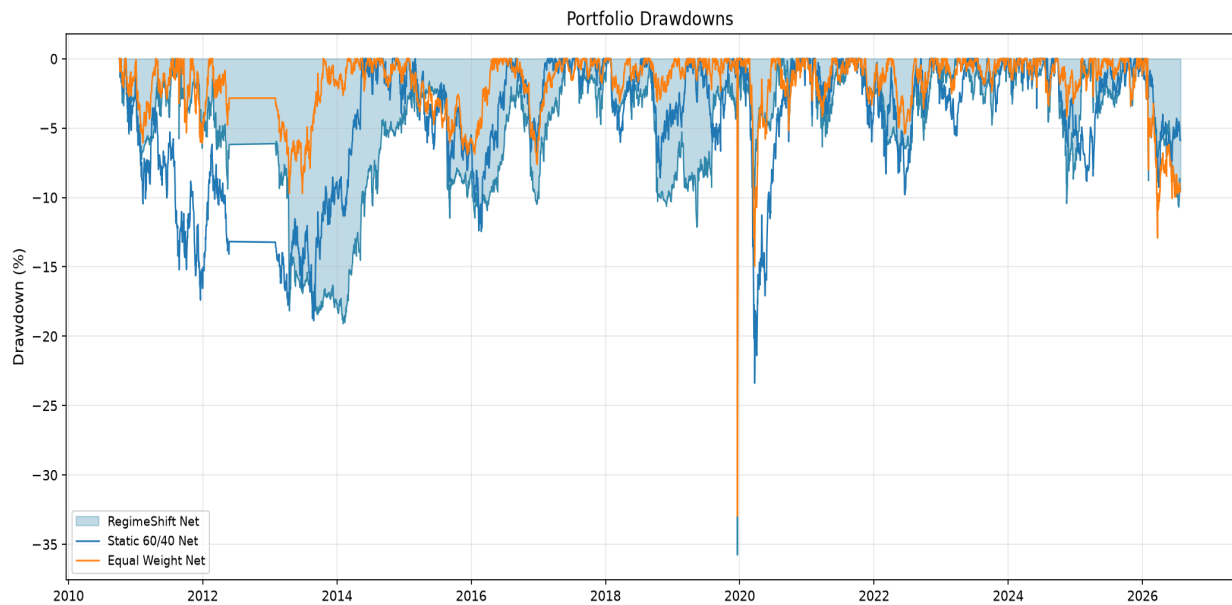
The equity leg is ^NSEI, an index-level return and signal proxy rather than a directly executable fill. GOLDBEES is the gold proxy. LIQUIDBEES is a short-duration liquid-bond/cash-equivalent proxy, not a sovereign bond or long-duration hedge.

Canonical data contract

The canonical dataset contains 4,090 source rows from 2010-01-04 to 2026-07-27. Its portable, line-ending-normalized SHA-256 is bound in the official manifest. The common evaluation horizon has 3,732 observations.

NEXT_CLOSE execution

Existing holdings earn the completed close-to-close return. Only afterwards may the model observe close-time data, select targets, trade, and pay costs. New targets first earn the next close-to-close return. This avoids same-bar execution.



Features and train-only scaling

Feature engineering and standardization are fitted on rolling training windows available at the decision time. No full-sample scaling or backward-filled signal is used in the official experiment.

Three-restart Gaussian HMM

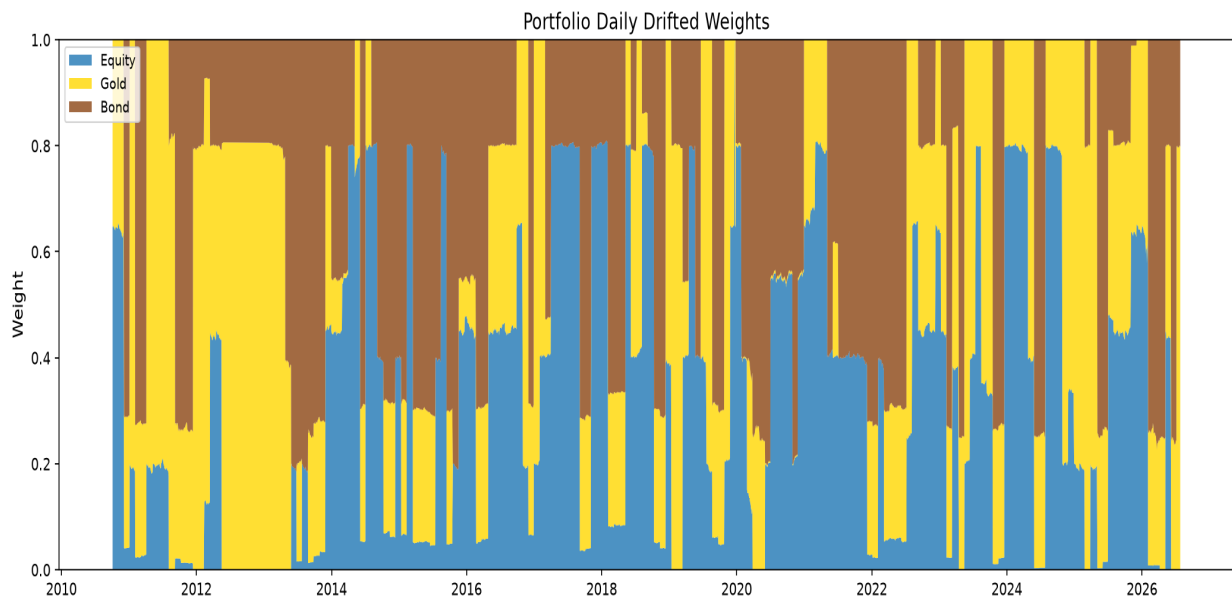
At every rebalance, three deterministic Gaussian HMM candidates are fitted. Candidate selection is based on training likelihood, never later investment performance.

State mapping and diagnostics

Restart diagnostics report 178 HMM fits and 534 converged candidates with no non-converged fallback. State labels are diagnostic interpretations, not forecasts or tradable states.

CVXPY portfolio construction

The full policy uses a constrained, long-only CVXPY optimizer conditioned on the selected state. Controlled policies only select targets; a shared engine retains the same return, drift, turnover, and cost logic.



Asset-level cost model

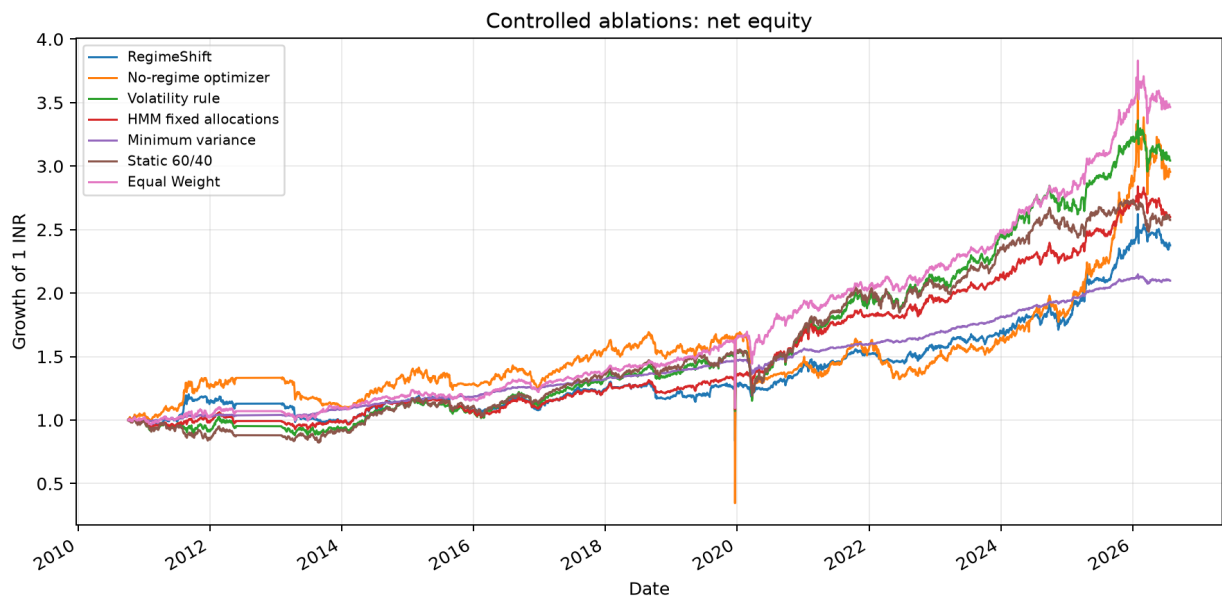
The official scenario is base: 10 bps one-way assumed cost for equity, gold, and defensive assets. The study does not model market impact, capacity, taxes, or realized bid-ask execution.

Official common-horizon results

RegimeShift: 6.02% CAGR, 0.400 Sharpe, 35.77% maximum drawdown. Static 60/40: 6.60%, 0.697, 23.39%. Equal weight: 8.75%, 0.570, 32.96%.

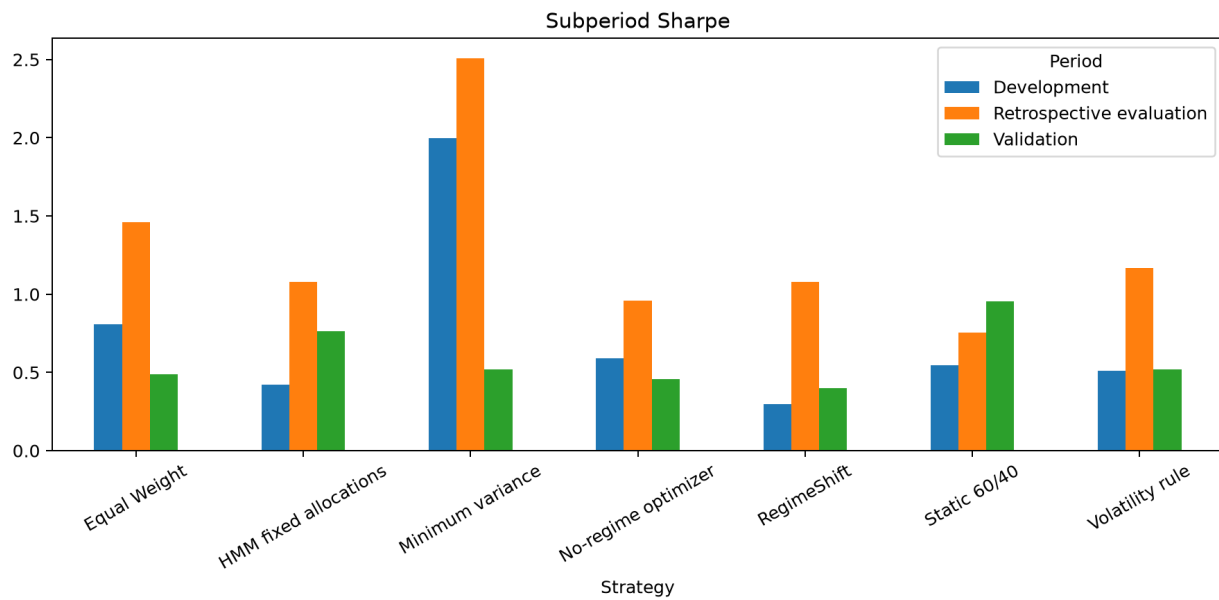
Controlled ablations

The HMM improves Sharpe over the no-regime optimizer (0.400 versus 0.280) and reduces drawdown. However fixed HMM allocations (0.635 Sharpe) and a transparent volatility rule (0.532 Sharpe) outperform the full optimizer on historical Sharpe.



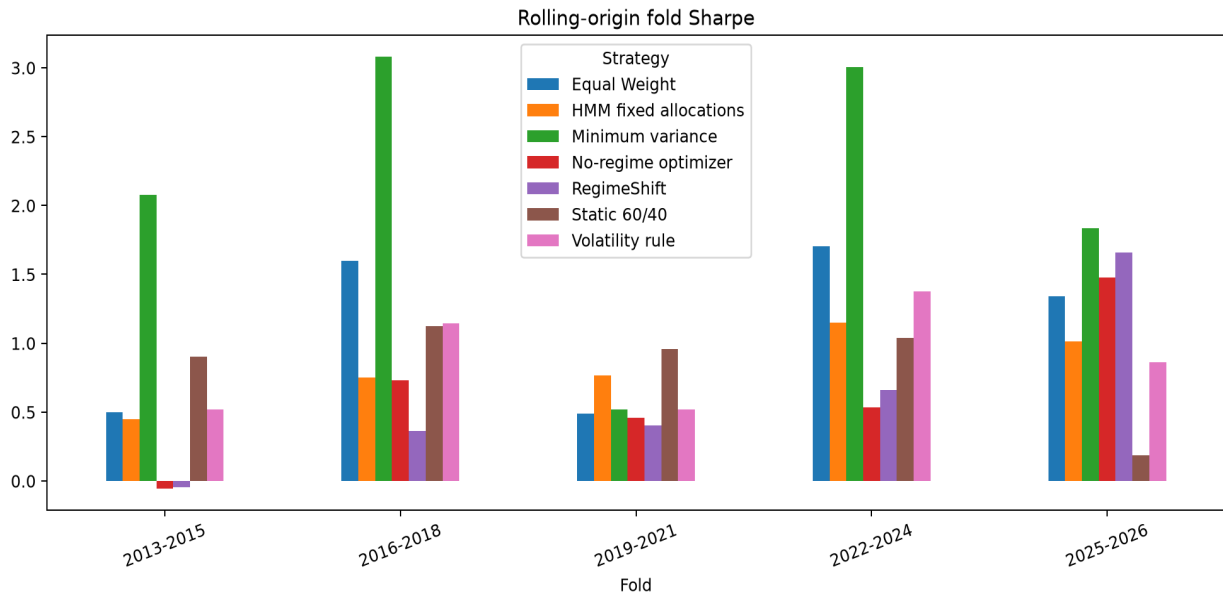
Chronological evaluation

Development, validation, and 2022-2026 retrospective summaries are reported separately. The last period is not described as a pristine holdout because the complete historical sample had been inspected during earlier development.



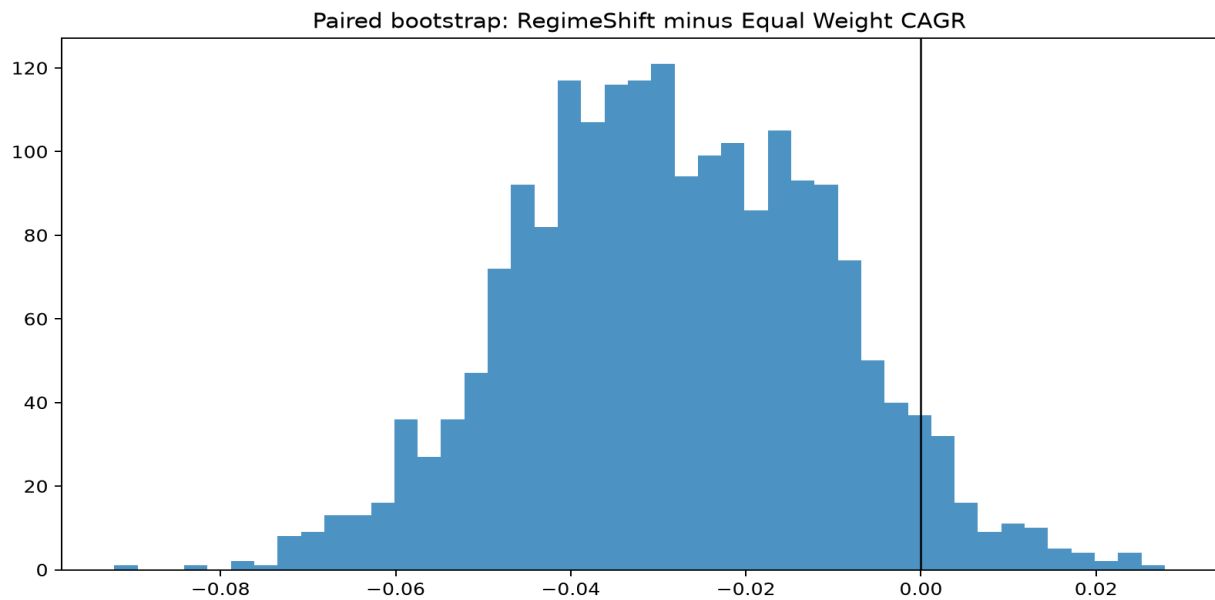
Rolling-origin evaluation

Five chronological folds summarize concentration and instability. Each fold reports only dates through its end, while the underlying NEXT_CLOSE decisions use no future observations.



Paired moving-block bootstrap

Confidence intervals use 21-trading-day blocks, 2,000 samples, seed 42, and paired date-block positions. The RegimeShift minus 60/40 Sharpe interval includes zero; this report makes no statistical-significance claim.



Negative findings

The official strategy does not beat either static benchmark on headline Sharpe. Complexity raises turnover, and simpler alternatives can compare favorably. These observations are retained rather than filtered from the release.

Limitations

This is retrospective research. ^NSEI is not directly tradable; capacity, impact, taxes, borrowing, product availability, and ETF tracking differences are outside scope. Results do not establish live trading performance.

Reproducibility

Run ``python scripts/run_research_suite.py``, build and execute the notebook, then run the test suite and ``python scripts/verify_release.py``. The official result lives in `results/submission` and research artefacts live in `results/research`.